

The Craft of Computing

What is this course about?

- First course in the Information Sciences major and minor
- Hands-on exploration of computing across three facets:
 - Command line interface
 - Python programming
 - Microcontrollers (ESP32)
- Goal: move from passive tech consumers to active creators and problem solvers
- Emphasis on teamwork, self-directed learning, and getting to know classmates

Course Objectives & Units

Course Objectives

- **Execute** basic computing tasks by following guidance or consulting resources
- **Foster** an inclusive learning environment and responsible conduct
- **Examine & troubleshoot** commands and code
- **Design & implement** new applications through tinkering and thoughtful design
- **Introspect** on learning and know when to seek help

Course Units

1. Computing with the command line
2. Exploring Python
3. Interfacing with Microcontrollers

Resources

Appropriate resource materials will be uploaded on Moodle.

Class Schedule

Compulsory Sessions

Day	Time	Location	Remarks
Monday	08:30 am – 10:30 am 10:40 am - 12:30 pm	Sannati 112	
Tuesday	03:50 pm – 05:50 pm	Sannati 004/107	During August
Wednesday	08:30 am – 10:30 am	Sannati 112	From September onwards

Office Hours (Optional)

- Exact hours will be announced soon
- There will be plenty of opportunities during timetabled slots to seek instructor help
- It is your responsibility to seek academic support

Topics And Assessment Schedule

A class-by-class schedule will be made available on Moodle.

Let's Pause

Any questions?

Assessments

Assessments

Assessment Type	Number of Assessments	Weightage of Each	Contribution to Overall Score
In-class Reflections	2	4% + 7%	11%
Group Presentation	1	5%	5%
Group Projects	3	5% + 10% + 10%	25%
Activity Reports	Best 3	3%	9%
Test	1	20%	20%
Learning Journal	1	20%	20%
Class Participation	1	10%	10%

In-class Reflections

- Individual assessment
- Two reflections: start of course (week 2) and end (week 14 / assessment week)
- First: views on computing, what they hope to learn
- Second: revisit those views, reflect on challenges and how they were addressed

Group Presentation

- Group assessment
- Class time will be dedicated to prepare presentation

Group Projects

- Group assessment (groups of 3 or 4)
- One open-ended project per unit — 3 total
- Class time dedicated to work on the project and seek instructor help

Activity Reports

- Group assessments
- Based on in-class activities in lab sessions
- Best 3 contribute 3% each

Test

- Individual, in-class assessment
- conducted after unit 2 has been completed (week 11 or 12)
- Pen and paper assessment

Learning Journal

- Individual, online journal built from Markdown files using a static site generator
- Minimum 6 entries required:
 - 1 on the group presentation
 - 3 on the three group projects
 - 2 on troubleshooting during class activities
- Students may write more entries if they wish

Class Participation

- Assessed on facilitating collaborative learning, responsible use of facilities, and consistent engagement
- Proactively seeking academic support counts as active engagement

Let's Pause

Any questions?

Policies

Attendance Policies

- 100% attendance is expected; 80% attendance is the minimum threshold to be eligible for a passing grade
- No late arrivals
- You will be marked absent if you do not come on time after in-class breaks

Assessment Policies

- You are expected to submit all assessments
- You will not be eligible for a passing grade unless you have submitted at least 65% of the assessments (by weightage)
- No extensions
- If you miss an assessment or fail to submit before the deadline, you get a zero

Academic Integrity

- No plagiarism
- No copying from each other
- No sharing of answers
- Both provider and copier of answers get the same penalty
 - Minor violations: zero grade for the entire assessment
 - Egregious violations: U grade for the course

AI Policy

- OK to use judiciously, with caution and with attribution
- It is your responsibility to ensure correctness
- There may be long-term consequences to becoming dependent on AI
- Detailed AI Policy will be communicated shortly

Other Points

- Bring your laptop to every session
- Ensure your laptop has sufficient charge to last the class

Any questions?

Grade Boundaries

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Final Score	Grade
$\geq 95\%$	O
90 – 95%	A
80 – 90%	A-
70 – 80%	B
60 – 70%	C
50 – 60%	D
40 – 50%	E
$< 40\%$	U

Thank You
